

Lecture 21 – Clustering

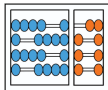
[ISLP 12.4; Pattern Classification 10.4, 10.9]

MC886 – Machine Learning

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Today's summary

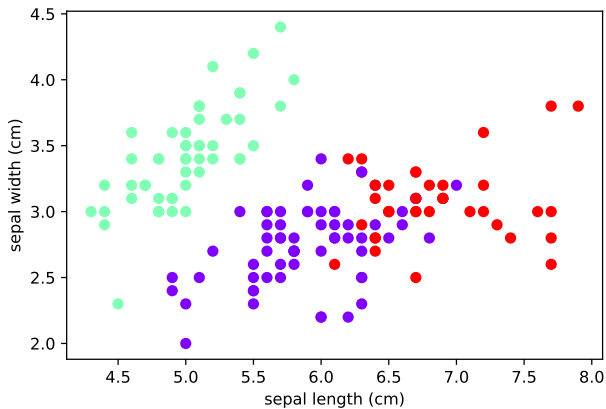
Recalling the unsupervised learning problem

The K -means clustering

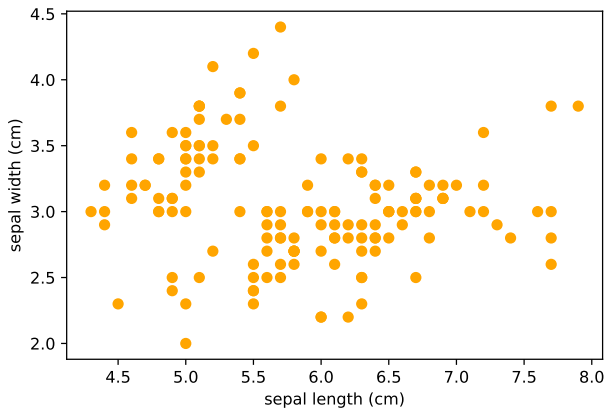
Hierarchical clusterings

Practical issues in clustering

supervised learning



Unsupervised learning



The Unsupervised Learning problem (another formulation)

- ❖ Given a random vector x that draws from $\mathcal{X}$ under a parameter vector θ , we want to estimate:

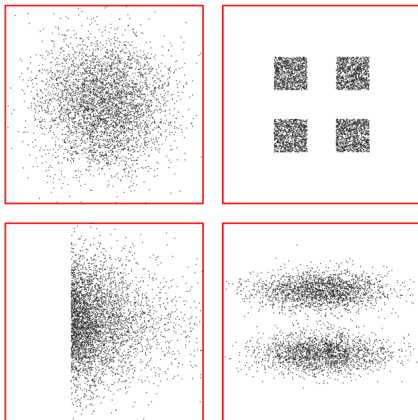
$$p(x|\theta)$$

- ❖ If we can assume the distribution shape of each element of θ , we can rely on parametric methods

For example, elements follow a normal distribution

- ❖ However, in practice, this is often not the case

When a normal distribution assumption fails



All examples have the same mean and variance!

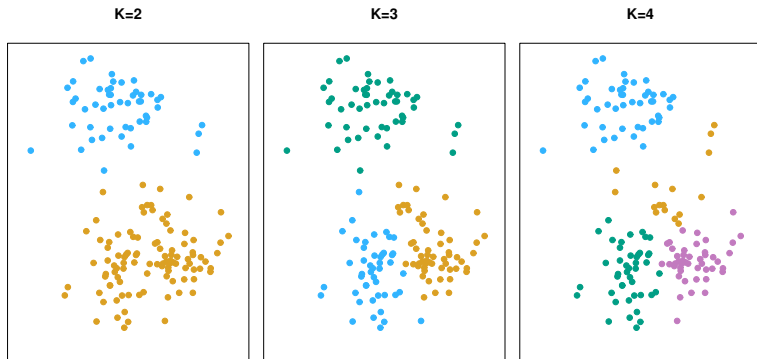
Clustering

A procedure that groups data points (observations) according to internal similarities within groups

How to measure similarity (or, equivalently, dissimilarity) between points?

- One possible measure: the distance between them
- For example, the Euclidean distance between two points

Clustering into K clusters (groups)



The K -means clustering

- ❖ A clustering algorithm that groups n points of a dataset into K disjoint clusters
- ❖ Let $C_1, \dots, C_K$ be sets containing the indices of points in each cluster, such that:
 1. $C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, n\}$
 2. $C_k \cap C_{k'} = \emptyset$ for all $k \neq k'$
- ❖ For example, if x_i is in the k th cluster, then $i \in C_k$

Key idea of K -means

- ❖ A *good* clustering is one with *within-cluster variation* as small as possible:

$$\min_{C_1, \dots, C_K} \left\{ \sum_{k=1}^K W(C_k) \right\}, \quad (12.15)$$

where $W(C_k)$ measures the amount by which points within the k th cluster differ from each other

- ❖ How to implement $W(C_k)$? One possibility is:

$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2, \quad (12.16)$$

which makes use of the squared Euclidean distance

Minimization of $W(C_k)$

- ❖ Plugging Eq. 10.16 into Eq. 10.15 yields:

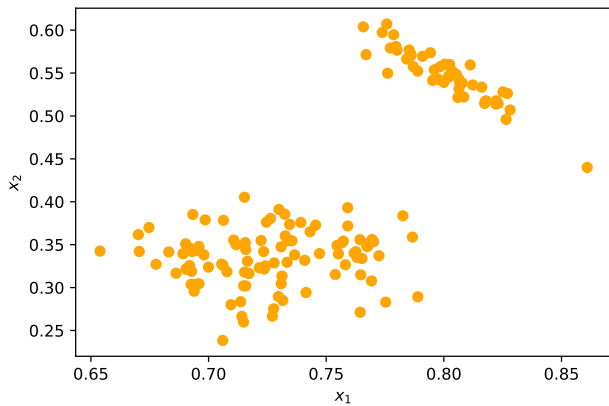
$$\min_{C_1, \dots, C_K} \left\{ \sum_{k=1}^K \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 \right\} \quad (12.17)$$

- ❖ Now we want to minimize Eq. 12.17
- ❖ However, there are K^n ways to partition n observations into K clusters!!! (this is a NP-hard problem 😞)
 - We'll seek then a suboptimum, yet satisfactory, solution

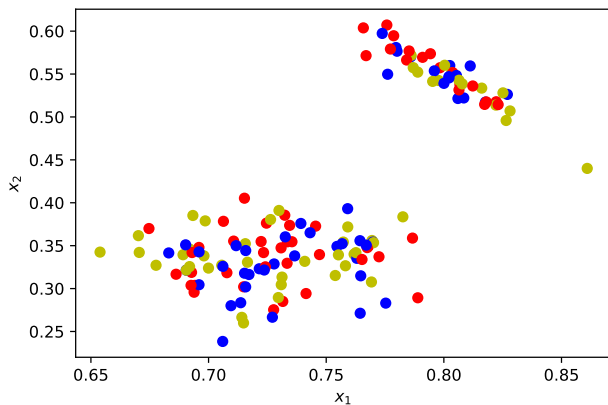
Algorithm 12.2. *K*-Means Clustering

1. Randomly assign a number, from 1 to K , to each of the observations. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
 - a) For each of the K clusters, compute the cluster centroid. The k th cluster centroid is the vector of the p feature means for the observations in the k th cluster.
 - b) Assign each observation to the cluster whose centroid is closest (where closest is defined using Euclidean distance).

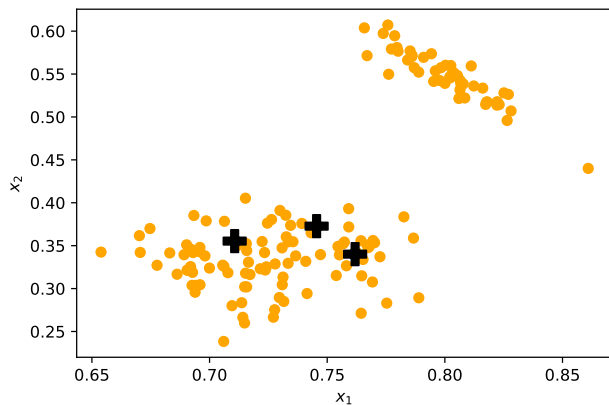
Simulation of K-MEANS



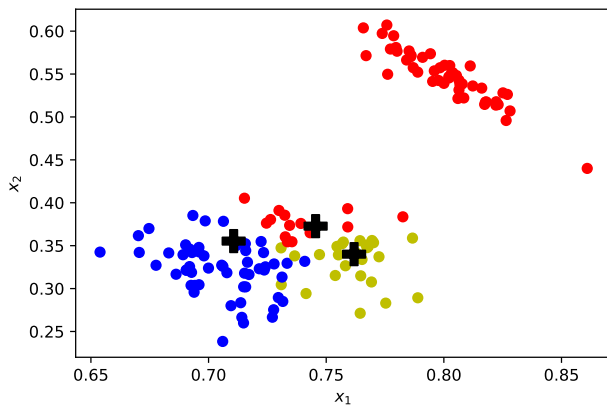
For each x_i , assign a random number in $\{1, \dots, K\}$ (step 1)



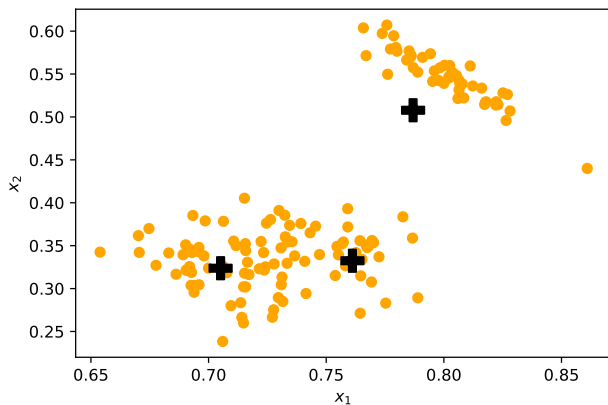
Centroid update step (step 2.a)



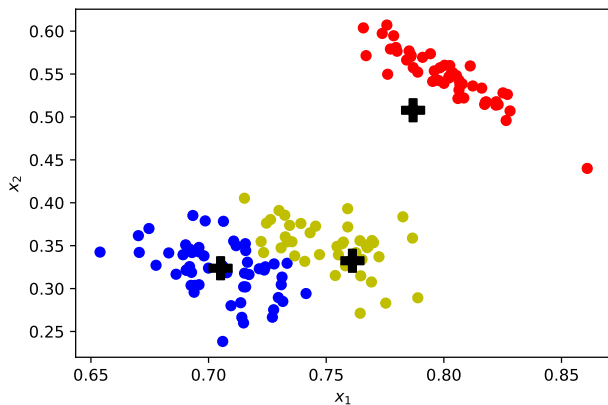
Cluster assignment step (step 2.b)



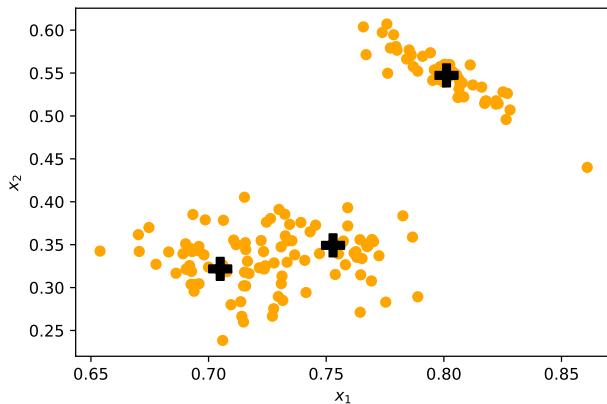
Centroid update step (step 2.a)



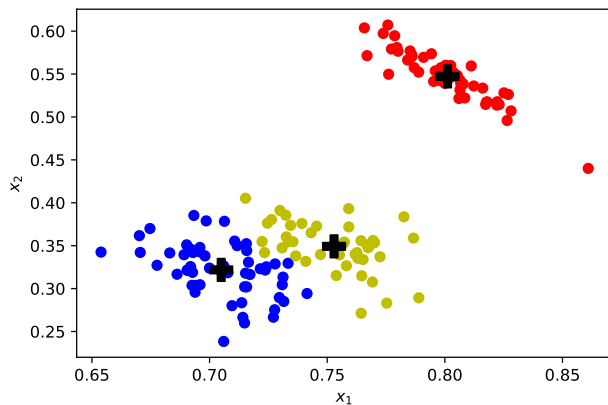
Cluster assignment step (step 2.b)



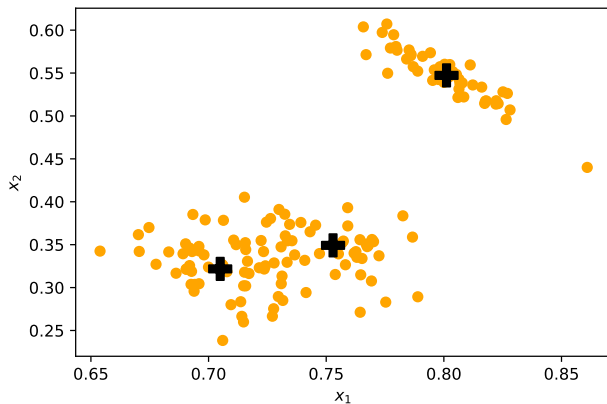
Centroid update step (step 2.a)



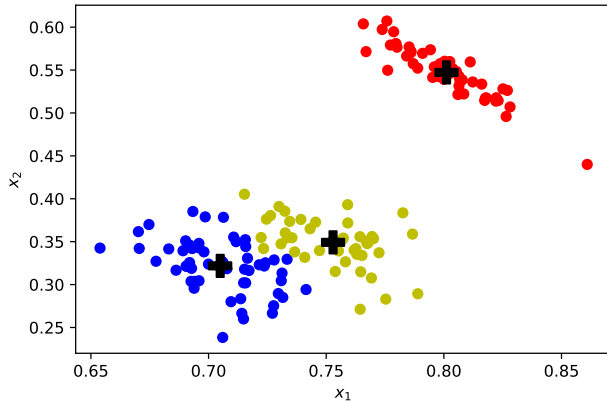
Cluster assignment step (step 2.b)



Centroid update step (step 2.a)



Cluster assignment step (step 2.b)



Convergence of K -means

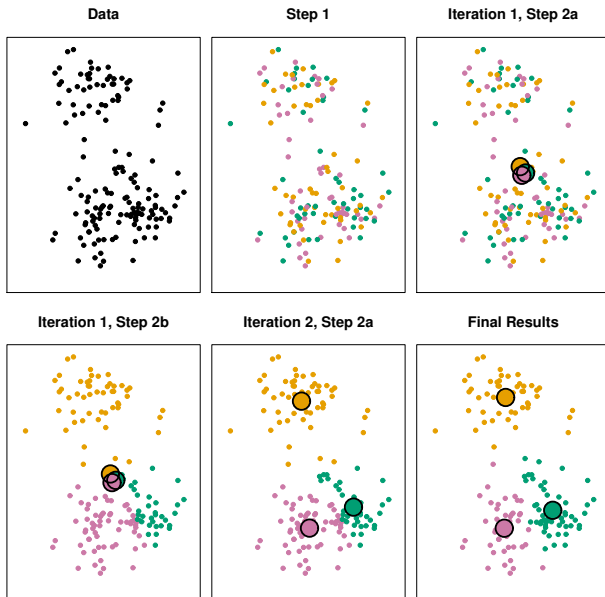
- ❖ At each step, the objective function (Eq. 12.17) is decreased.
Consider:

$$\frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{ij})^2, \quad (12.18)$$

where $\bar{x}_{ij} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}$ is the mean for variable j in cluster C_k

- ❖ In Step 2(a), the cluster means for each variable are the constants that minimizes the sum-of-squared deviations
- ❖ In Step 2(b), reassigning the observations to new clusters can only improve Eq. 12.18
- ❖ The objective function (Eq. 12.17) never increases

Another simulation of K -means



K-means results for 6 different initializations



Value above graphs indicates the respective final objective value

Choosing a value for K

- ❖ The “elbow method”
- ❖ Knowledge about your problem
- ❖ Using the silhouette method

Silhouette

- ❖ A method for assessing cluster consistency
- ❖ For a given point, it measures how similar it is to other points within the cluster (cohesion), and at the same time how dissimilar it is to points of other clusters (separation)
- ❖ It ranges from -1 to +1
- ❖ If most points have high values, then the cluster configuration is considered appropriate

Computing the silhouette $s(x_i)$ for a point x_i

- ❖ First, we calculate the mean intra-cluster distance:

$$a(x_i) = \frac{1}{|C_k| - 1} \sum_{i' \in C_k, i' \neq i} \text{dist}(x_i, x_{i'}),$$

where $|C_k|$ is the size of the cluster that has x_i , and dist can be the Euclidean distance. If $|C_k| = 1$ then $a(x_i) = 0$

- ❖ Second, we calculate the mean nearest cluster distance:

$$b(x_i) = \min_{\ell \neq k} \frac{1}{|C_\ell|} \sum_{x_{i'} \in C_\ell} \text{dist}(x_i, x_{i'}),$$

where $|C_\ell|$ is the size of a cluster that doesn't have x_i

Computing the silhouette $s(x_i)$ for a point x_i

- ❖ Finally, the silhouette for the point x_i is then given by:

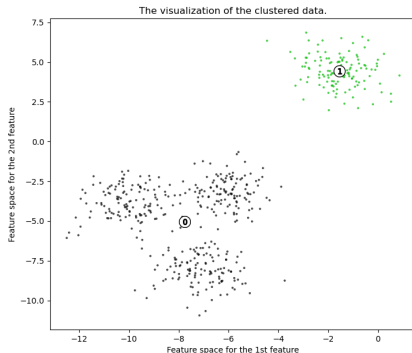
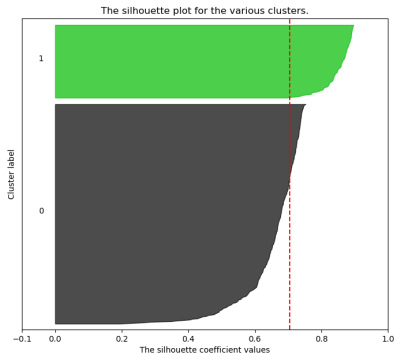
$$s(x_i) = \begin{cases} 1 - a(x_i)/b(x_i), & \text{if } a(x_i) < b(x_i) \\ 0, & \text{if } a(x_i) = b(x_i) \\ b(x_i)/a(x_i) - 1, & \text{if } a(x_i) > b(x_i) \end{cases}$$

- ❖ x_i close to:

- +1: it's far away from neighboring clusters
- 0: it's near the decision boundary between clusters
- -1: it might have been assigned to the wrong cluster

Selecting the number of clusters with silhouette

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 2$

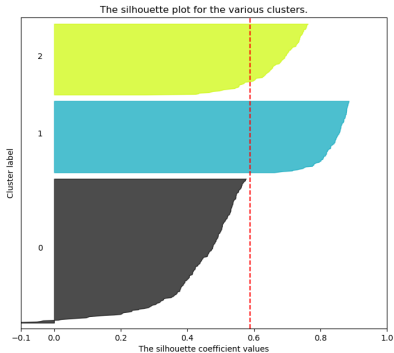


scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Dashed line is the average $s(x_i)$

Selecting the number of clusters with silhouette

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 3$

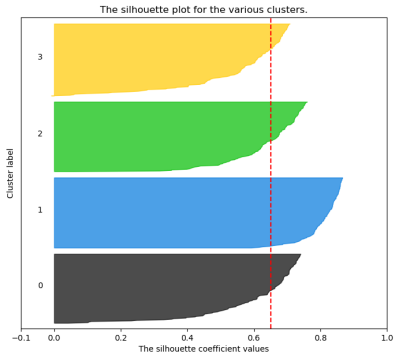


scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Dashed line is the average $s(x_i)$

Selecting the number of clusters with silhouette

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 4$

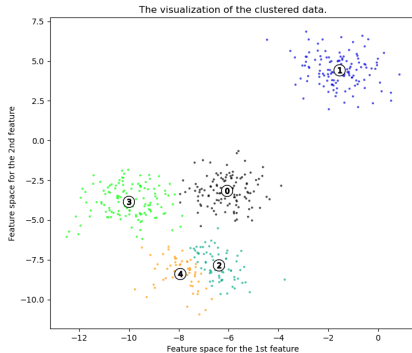
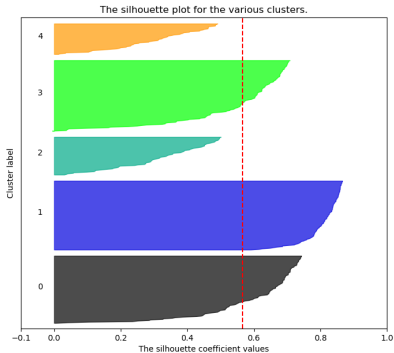


scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Dashed line is the average $s(x_i)$

Selecting the number of clusters with silhouette

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 5$

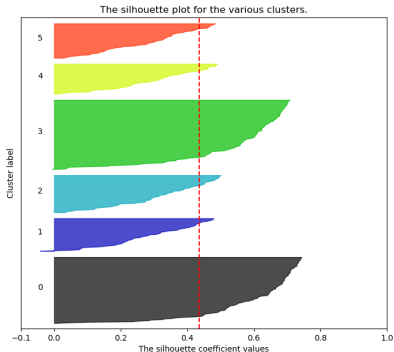


scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Dashed line is the average $s(x_i)$

Selecting the number of clusters with silhouette

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 6$



scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Dashed line is the average $s(x_i)$

Hierarchical data is everywhere!

- ❖ Up to now, we treated clusters as disjoint sets
- ❖ However, data is often organized in a hierarchical fashion
- ❖ **Examples:** taxonomy, geography, astronomy, biology, ...
- ❖ A suitable data representation to capture such hierarchies?

General principles for hierarchical clustering

- ❖ High-level principles for that procedure?

Idea 1: Group first pairs that are the closest to each other

Idea 2: Ungroup first pairs that are the farthest to each other

- ❖ **Two approaches:** agglomerative and divisive

Agglomerative clustering formalization

- ❖ Consider a sequence of partitions of the n points into clusters
- ❖ First partition contains n clusters, each one with one point
- ❖ The second partition contains $n - 1$ clusters
(one cluster with two points, all others with a single one)
- ❖ The i th partition contains $n - i + 1$ clusters

Example: clustering a dataset with $n = 8$ points $(x_1, \dots, x_8)$

$i = 1:$ $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$

$i = 2:$ $x_1, x_2, x_3, x_4, x_5, \{x_6, x_7\}, x_8$

$i = 3:$ $x_1, \{x_2, x_3\}, x_4, x_5, \{x_6, x_7\}, x_8$

$i = 4:$ $x_1, \{x_2, x_3\}, \{x_4, x_5\}, \{x_6, x_7\}, x_8$

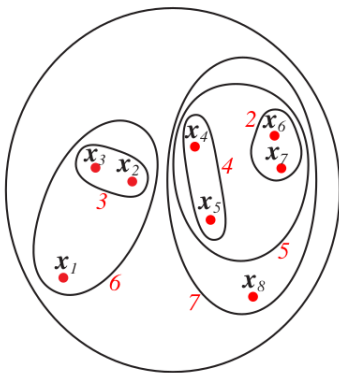
$i = 5:$ $x_1, \{x_2, x_3\}, \{\{x_4, x_5\}, \{x_6, x_7\}\}, x_8$

$i = 6:$ $\{x_1, \{x_2, x_3\}\}, \{\{x_4, x_5\}, \{x_6, x_7\}\}, x_8$

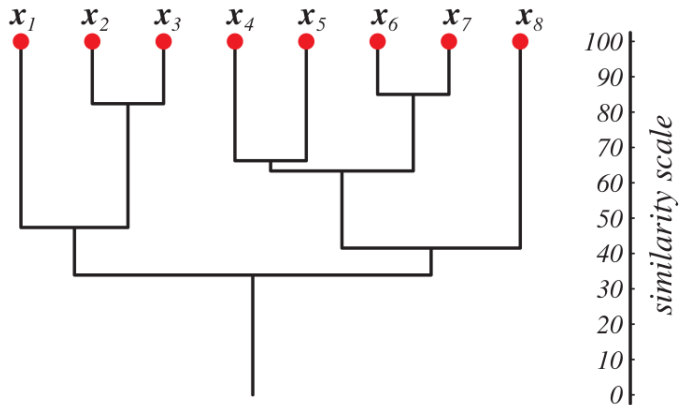
$i = 7:$ $\{x_1, \{x_2, x_3\}\}, \{\{\{x_4, x_5\}, \{x_6, x_7\}\}, x_8\}$

$i = 8:$ $\{\{x_1, \{x_2, x_3\}\}, \{\{\{x_4, x_5\}, \{x_6, x_7\}\}, x_8\}\}$

Venn diagram representation



Dendrogram visualization



Implementing an agglomerative clustering

Basically, we need two things:

1. A distance metric
2. An agglomeration criterion

Some distance metrics

❖ Euclidean:

$$\text{dist}_{\text{Euclidean}}(x_i, x_{i'}) = \sqrt{(x_{i1} - x_{i'1})^2 + \dots + (x_{ip} - x_{i'p})^2} = \|x_i - x_{i'}\|$$

❖ Jaccard (also known as *Jaccard index*):

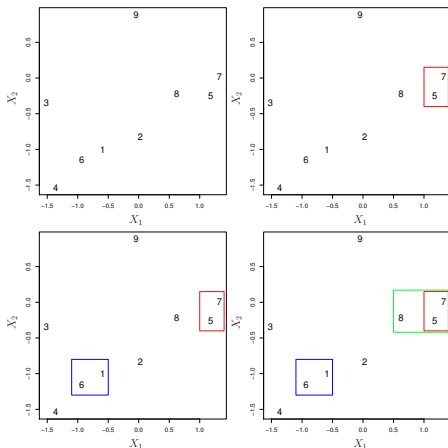
$$\text{dist}_{\text{Jaccard}}(x_i, x_{i'}) = \frac{|x_i \cap x_{i'}|}{|x_i \cup x_{i'}|} = \frac{|x_i \cap x_{i'}|}{|x_i| + |x_{i'}| - |x_i \cap x_{i'}|}$$

- ❖ The Jaccard index is equivalent to the *Intersection over Union* (IoU) in the Image Processing field

Common agglomeration criteria

Linkage	Description
Complete	Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the largest of these dissimilarities.
Single	Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the smallest of these dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one-at-a-time.
Average	Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the average of these dissimilarities.

Example of using the complete linkage criterion (with Euclidean distance)

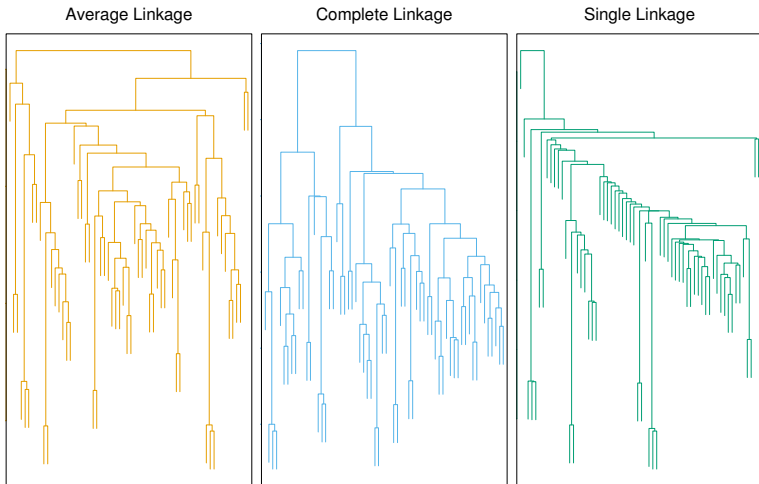


Top right: $\{5\}$ and $\{7\}$ are the closest trivial clusters and are merged together

Bottom left: $\{6\}$ and $\{1\}$ are the closest trivial clusters and are merged together

Bottom right: $\{8\}$ and $\{5, 7\}$ are the closest trivial clusters and are merged together

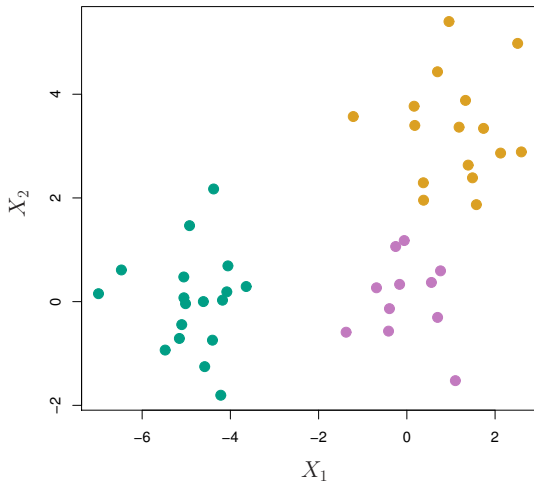
Different linkage criteria yield different clusters



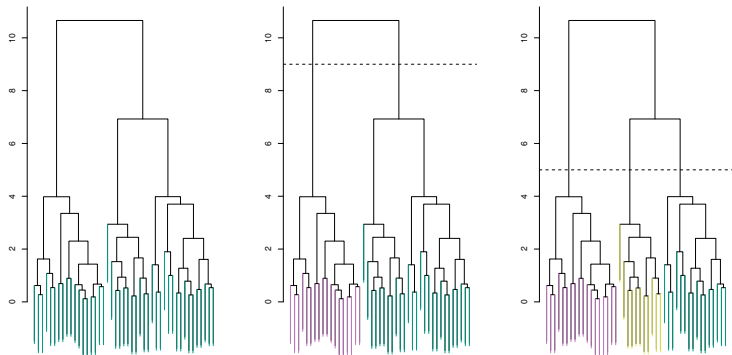
Algorithm 12.3: *Hierarchical Clustering*

1. Begin with n observations and a measure (such as Euclidean distance) of all the $\binom{n}{2} = n(n-1)/2$ pairwise dissimilarities. Treat each observation as its own cluster.
2. For $i = n, n-1, \dots, 2$:
 - (a) Examine all pairwise inter-cluster dissimilarities among the i clusters and identify the pair of clusters that are least dissimilar (that is, most similar). Fuse these two clusters. The dissimilarity between these two clusters indicates the height in the dendrogram at which the fusion should be placed.
 - (b) Compute the new pairwise inter-cluster dissimilarities among the $i-1$ remaining clusters.

Agglomerative hierarchical clustering example

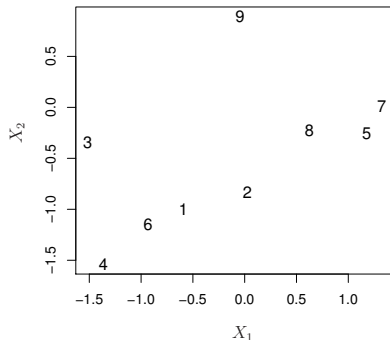
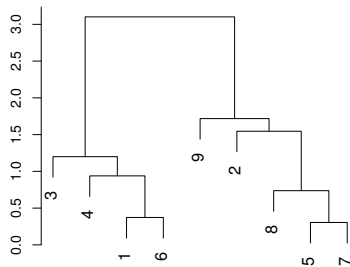


Agglomerative hierarchical clustering example



Dashed lines indicate possibilities of “cut” in the dendrogram to yield different number (K) of clusters

Another example of hierarchical clustering



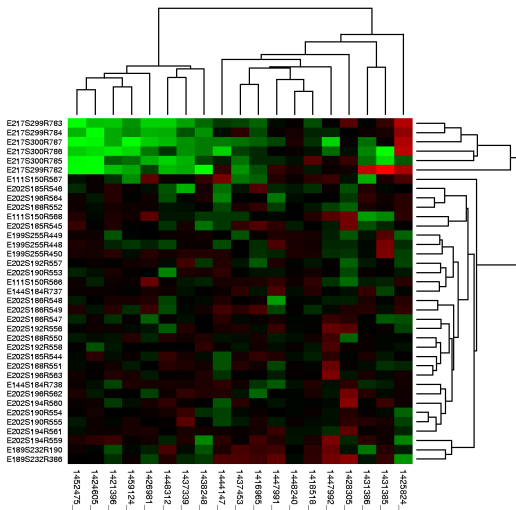
Left: Dendrogram generated by an agglomerative hierarchical clustering with Euclidean distance and complete linkage

Right: Data used to generate that dendrogram

Clustering of a $n \times p$ data matrix

- ❖ If we organize a set of p -dimensional n points into a $n \times p$ matrix, we can carry out two hierarchical clustering procedures
- ❖ First, we cluster the n observations (rows)
- ❖ Then, we apply hierarchical clustering in the *transpose matrix* (i.e., in this clustering, we have p points, each of them with dimensionality n)
- ❖ The yielded result often is better visualized through a *heatmap*

Example of heatmap with two hierarchical clusterings



Public domain image from Wikipedia.

Divisive hierarchical clustering

- ❖ “Top-down” approach instead of a “bottom-up” one
- ❖ At each iteration, we need to find the best partition among the clusters
- ❖ Computational complexity of that procedure?



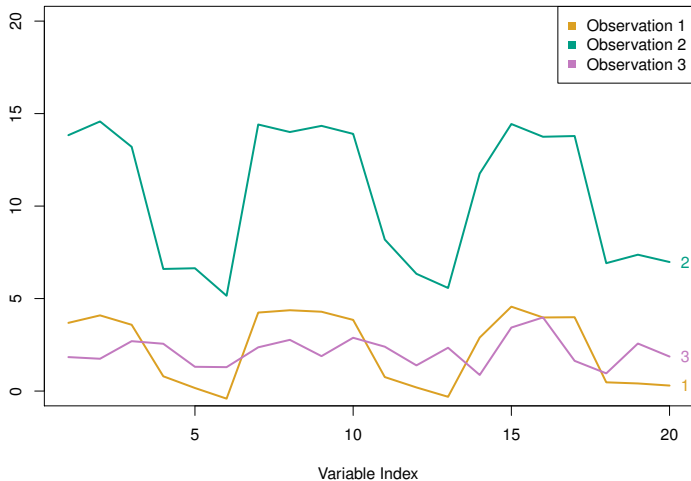
Time complexity of different methods

- ❖ K -means: $O(Kn) = O(n^2)$
(provided p and number of iterations are constants; if K is also a constant, the algorithm is asymptotically linear!)
- ❖ Agglomerative clustering: $O(n^3)$ (too slow for large datasets)
- ❖ Divisive clustering: $O(2^n)$ (without heuristics)
 - Possible heuristic: run K -means (with $K = 2$) on a cluster to split it into two disjoint parts

Practical issues clustering

1. There is no “correct choice” of method or hyperparameters (though some choices might make more sense according to the knowledge about your dataset)
2. Clustering procedures are often sensitive to outliers
3. Choice of clustering distance is also context-dependent (e.g., sometimes it might be more reasonable to look for *correlations* among points)
4. Scaling features might affect clusters

3. Correlation distance vs. Euclidean distance



4. Scaling features might affect clusters

